

SIGNAL ANALYSERS

Signal Analysers	Company	Features	Application
MS933A PXIe performance vector signal analyser 	Keysight	-168dBm/Hz displayed average noise level (DANL) with preamplifier and noise corrections - Frequency extension to 50GHz (minimum frequency of 3.6GHz) with option FRX 40MHz (standard), 100MHz or 160MHz analysis bandwidth, and optional wide-band IF output with option WB1	- Aerospace and defence manufacturing and depot test - Wireless device design validation and manufacturing
R&S FSW85 signal analyser 	R&S	- Frequency range: 2GHz to 85GHz - Low phase noise: -137dBc (1Hz) at 10kHz offset (1GHz carrier) - Up to 2GHz analysis bandwidth - Real-time analysis up to 160MHz bandwidth - High-resolution 31cm (12.1-inch) touchscreen for convenient operation - Multiple measurement applications can be run and displayed in parallel	Ideal for analysing radar systems

SPECTRUM ANALYSERS

Spectrum Analysers	Company	Features
Spectrum Rider FPH handheld spectrum analyser 	R&S	- Frequency ranges from 5kHz to 2/3/4GHz; upgrade via keycode - Eight-hour battery life - Measurement accuracy of typically 0.5dB between 10MHz and 3GHz
TE-3000 	Antenna Technology Communications Inc. (ATCI)	- Frequency margin from 5MHz to 1000MHz (terrestrial), from 950MHz to 2150MHz (satellite) - Button and touchscreen control - More than eight-hours battery time - Compatible with analogue, digital (DVB) satellite standard
PSA Series 2 (PSA1302, PSA2702) 	Aim Tti	- Frequency PSA1302: 1MHz - 1300MHz, PSA 2702: 1MHz - 2700MHz - USB interfaces for flash drives and PC connection - Resolution bandwidth (RBW): 15kHz, 280kHz, 1MHz or auto - Button and touchscreen control
PSA Series 5 (PSA3605, PSA6005) 	Aim Tti	- Frequency PSA3605: 10MHz - 3600MHz, PSA6005: 10MHz - 6000MHz - USB interfaces for flash drives and PC connection 300Hz to 10MHz (1:3:10 sequence) or auto - Button and touchscreen control
RTSA7500 	Berkeley Nucleonics	- Frequency range from 100kHz to 8/18/27GHz - Real-time bandwidth (RTBW) up to 100MHz - Real-time FPGA triggering for detection of elusive, time-varying signals
RF Explorer 6G Combo 	RF Explorer	- Frequency range from 15MHz - 2700MHz, 4850MHz - 6100MHz - Pocket size and lightweight - Frequency resolution of 1.0KHz - Average noise level of 0.5dBm

analysers much more exciting than the previous generation is their easy integration within a framework of a

computer. Most new instruments are supported by powerful application software, which can run on various

operating systems. In a sense, this feature enables the instruments to form an analysis network that can be controlled from a central post."

We are now quite rooted in the era of touchscreens for test equipment, with newer equipment coming with touchscreens as big as 36 centimetres. These screens have also enabled an era of user interface (UI) improvements.

While engineers previously had to click through and dig deep into menu structures to reach specific functions, Gupta claims that newer Keysight equipment allows him or her to modify something with three clicks or less. "Better UI also enables enhanced pulse-measurement capabilities for white-band radar and electronics warfare applications," he adds.

Another support which is now available is the possibility to obtain any movement, location and alignment. "This is ideal for automatic measurements. Such instruments feature a gyroscope and acceleration sensor as well as a digital compass and an optional integrated global position system," adds Sharma.

Support for newer standards and markets

A recent trend in the radio frequency (RF) spectrum analyser market is to provide integrated test solutions that feature many